MILITARY SPECIFICATION SHEET

STARTER, AIR TURBINE, ACCESSORY DRIVE COMBINATION,
AIRCRAFT ENGINE, MODEL ATS/AD-3

This specification sheet is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the starter described herein shall consist of this document and the latest issue of Specification MIL-S-22518(AS).

REQUIREMENTS

Applicable paragraphs of MIL-S-22518A(AS)	
3.8.1	Envelope: Figure 1
3.7	Wt, Max (Lb); 33
4.5.19	Aircraft Model: EA-6B
4.5.20	Engine Model: J52
3.8.2.1	Accessory Drive: Engine: Figure 1 pad "B": Figure 1
3.5.5 3.9.1	Rotation Viewed from Pad "B" End: Pad "A": Clockwise Pad "B": Counterclockwise
3.3.2	Ambient Temperature Operating Range (°F): Starter: -65 to +160 Accessory Drive: -65 to +250
3.3.2	Exposure Temperature Range (°F): -100 to +275
3.3.2.2 3.5.7	Control Valve: Opening Rate, Max: 22 psig per second pressure rate for first 0.5 second; full open in 3 to 5 seconds Closing Rate, Max: 0.5 second

FSC 2995

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REQUIREMENTS (Continued)
Applicable paragraphs
of MIL-S-22518A(AS)

3.5.1 Rated Conditions:

Starter (Pad "A"):
Drive Speed (RPM): 1020
Drive Torque, Min (Lb-Ft): 202
Airflow, Max (Lb per Min): 90
Inlet Air Total Pressure (In. Hg A): 100
Inlet Air Total Temperature (°F): 345
Exhaust Gas Discharge Pressure (In. Hg A): 29.9

Accessory Drive (Pad "B"): The unit shall have a continuous rating of 55 HP, overload ratings of 65 HP for one hour, 82 HP for two minutes and 110 HP for five seconds and fault clearing of 140 HP. The 55 HP, 65 HP, and 82 HP ratings shall apply through a speed range of 3500-7200 RPM. The 110 HP rating shall apply through a speed range of 4000-7200 RPM. The accessory drive output RPM shall be the same as the starter output RPM.

3.5.1 cutout RPM: 3200 plus 200 minus 0

3.5.2 Maximum Temperature Limit: The air turbine shall be
3.5.8 capable of operating with a maximum inlet air total temperature limit of 650°F at rated inlet air total pressure.

3.5.8 Altitude: The ATS/AD shall not be damaged when subjected
4.5.17 to 75,000 ft altitude in the overrunning mode of operation.

3.5.9 Attitude: The ATS/AD shall be capable of meeting the operational requirements when the ATS/AD centerline, taken through the pad "A" shaft axis, is in any position within 10 degrees of the horizontal position. Inverted flight test time is 15 minutes.

3.5.10 Overrunning (Qualification) : The ATS/AD shall be overrun
4.5.10 at a 7250 RPM pad "A" drive speed for a total of 20 hours. Testing need not be continuous, but test periods shall not be less than 5 hours.

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REQUIREMENTS (Continued)
Applicable paragraphs
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3.5.10 Overrunning (Acceptance): The ATS/AD shall be overrun
4.5.10 at a Pad "A" speed of 7200 RPM and subjected to a Pad "B" load of 55 HP for 5 minutes.

4.5.4 Initial Calibration:

Pad "A" Drive

Torque (Lb-Ft)	Speed RPM	Speed Condition
270 Min, 385 Max	0	Stall
202 Min ---	1020	Rated
25 Min ---	3200 to 3400	cutout

Airflow, Max (Lb per Min)	Speed RPM	Speed Condition
90	0	Stall

4.5.5 No-Load Operation (Qualification):
Time (Second): 60
Pad "A" Drive Speed (RPM), Min: 3560

4.5.5.1 Burst Test:
Inlet Air Total Pressure (PSIA): 75
Inlet Air Total Temperature (°F): 455 ±10°F

4.5.6 No-Load Operation Acceptance): pad "A"
Drive Speed (RPM), Min: 3560

3.5.5 Endurance Test, Accessory Drive. The unit shall be mated with a 30 KVA constant speed drive and generator and mounted in a reamer dynamically similar to its mounting in the aircraft. This phase of the endurance testing need not be continuous; however, test periods shall not be less than 5 hours duration.

The complete package shall be tested for a total of 460 hours per the following schedule:

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Applicable paragraphs
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Generator Load % of rated	Input Speed RPM	Operating Time Hours
10	3500	3.5
10	5300	8.0
10	7200	11.5
50	3500	10.5
50	5300	24.0
50	7200	34.5
100	3500	58.5
100	5300	136.0
100	7200	173.5

During the test, 150% rated generator load shall be applied for 2 minutes every hour. 200% rated generator load shall be applied for five seconds every four hours at a speed greater than 4000 RPM.

The complete package shall be cycled for a total of 40 hours per the following schedule.

Generator Load % of Rated	Operating Time Minutes
10	5
50	5
100	5
10	5
100	5
50	5

The above cycling shall be accomplished at speed of 3500, 4000, 4500, 5000, 5500, 6000, 6500, 7000, 7200 RPM.

Signs of structural weakness or wear incurred during the accessory drive endurance test that will interfere with continued operation shall be cause for rejection.

Endurance Test, Accessory Drive High Torque Test. At an input speed of 3500 RPM, the unit shall be cycles for 15 minutes at 55 HP and 2 minutes at 82 HP for a total of two hours.

4.5.7 Endurance Test Starter: No. of cycles: 1200. Each cycle shall consist of three consecutive phases as follows:

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REQUIREMENTS (Continued)
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Test Stand Rotor Polar Moment of Inertia	Phase A - Acceleration		Phase B - At Following Conditions			Phase C - Acceleration	
	Drive Speed*	Time Max	Drive Speed*	Drive Torque Min	Time	Drive Speed* Min	Time
lb-Ft^2	RPM	Sec	RPM	Lb-Ft	Sec	RPM	Sec
430	1200	8	1020	202	10	2350	12

*Pad "A"

NOTE : The above values do not allow for flywheel windage losses, test stand drag, or inertia of the ATS/AD. In order to determine compliance with this specification, the test equipment shall be calibrated and specific test results corrected accordingly. If the acceleration specified for phase C is not developed because of test stand drag torque, the test stand inertia load may be reduced or external power may be applied to assist acceleration to cutout speed. Time for end of phase C to cutout shall not exceed 34.5 seconds.

4.5.9 Rated Torque Calibration:
Pad "A" Drive Speed (RPM), Min: 1020
Airflow, Max (Lb per Min): 90

4.5.10 Overrunning: The ATS/AD shall be overrun at a pad "A" drive speed of 7250 RPM for a total of 20 hours. Testing need not be continuous, but test periods shall not be less than 5 hours.

4.5.11.1 Low Temperature Operation:
Starter: The inlet air and the stabilized ATS/AD temperatures for the low temperature test shall be as follows:

Inlet Air Total Pressure In. Hg Abs ±0.5	Inlet Air Total Temperature °F ± 10	ATS/AD Temperature °F (Maximum)
140	260	-65

After the ATS/AD temperature has been stabilized for a minimum of 8 hours at -65°F, three consecutive cycles at -65°F ambient air temperature shall be run. Each cycle shall consist of the following phases:

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REQUIREMENTS (Continued)
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Phase A: The ATS/AD shall accelerate the flywheel having a polar moment of inertia of 430 lb-ft^2 for a period of 8 seconds from rest.

Phase B: After a period of 8 seconds from rest, a torque load shall be applied to maintain the pad "A" drive speed at a constant value (±100 RPM) for a period of 10 seconds.

Phase C: At the end of phase B the torque load shall be removed and the ATS/AD allowed to accelerate the flywheel to cutout speed or for an additional period of 12 seconds, whichever occurs first. Upon completion of the 12-second period or phase C if cutout speed has not been obtained, the ATS/AD shall be motored by an external source to a speed that will cause the automatic shutoff controls to operate.

Accessory Drive: After the ATS/AD temperature has been stabilized for a minimum of 8 hours at -65 F, the ATS/AD shall be operated at a -65°F ambient air temperature at 3500 RPM input at Pad "A" for a period of 5 hours with a 55 HP continuous load applied at pad "B," and 7200 RPM for 5 hours with a 55 HP continuous load applied at pad "B." An 82 HP load shall be applied for 2 minutes once during this test at each of the above speeds. A 110 HP load shall be applied for 5 seconds twice during this test at 4000 RPM and 7200 RPM.

4.5.11.2 High Temperature Operation:
Starter: The inlet air pressure and temperature and the stabilized ATS/AD temperatures for the high temperature test shall be as follows:

Inlet Air Total Pressure In. Hg Abs ±0.5	Inlet Air Total Temperature °F ±10	ATS/AD Temperature 'F (Minimum)
90	650	160

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REQUIREMENTS (Continued)
Applicable paragraphs
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After the ATS/AD temperature has been stabilized for a minimum of 8 hours at +160°F, three consecutive cycles at a +160°F ambient air temperature shall be run. Each cycle shall be as specified for phases A, B and C of 4.5.11.1. Accessory Drive: After a minimum 8-hour soak period at an ambient air temperature of +250°F, the ATS/AD shall be operated with ambient air at 250°F minimum at 3500 RPM speed at pad "A" drive for a period of 5 hours and at 7200 RPM for 5 hours with a 55 HP continuous load applied to the accessory drive output. An 82 HP load shall be applied for 2 minutes once during the test at each of the above speeds. A 110 HP load shall be applied for 5 seconds twice during this test at 4000 RPM and 7000 RPM.

4.5.12 Consecutive Cycle Operation: No. of cycles: 5

4.5.14 Vibration: All vibration testing shall be conducted with the specified accessory or a dummy, having a weight of 17.4 pounds and a center of gravity 4 inches from the mounting face, mounted on the accessory pad (pad "B"). Mounting shall be dynamically similar to that encountered in service use. Vibration tests shall be conducted in accordance with MIL-STD-810, Method 514, Procedure I, Curve F (Parts 1, 2 and 3). In addition, the ATS/AD shall be capable of withstanding a loading of 10g bending moment applied at the ATS/AD e.g. in any direction without structural damage.

4.5.18 pad "A" Drive Shear Section Strength: Torque (Lb-Ft): 535 to 620

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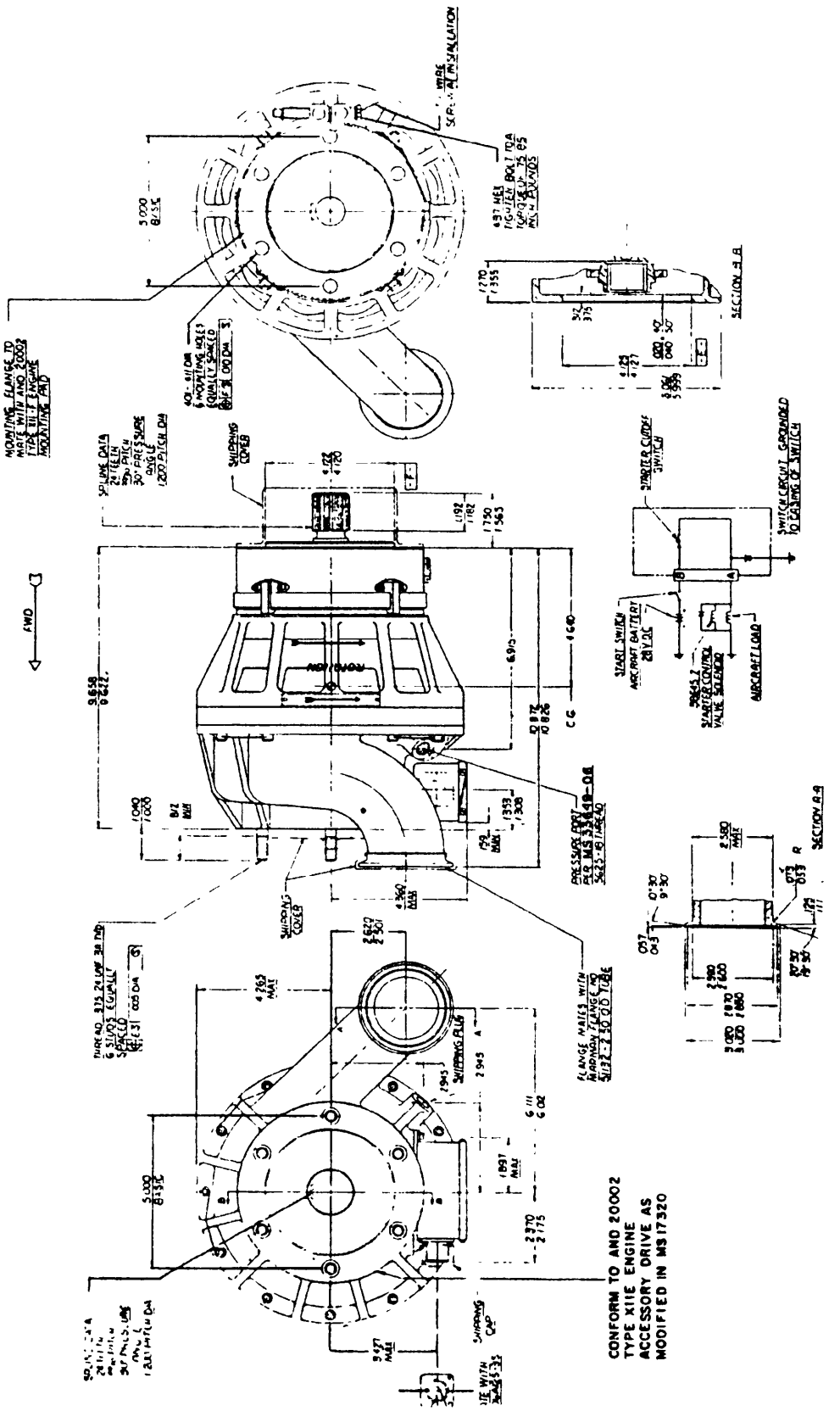


FIGURE 1. Starter, Air Turbine, Model ATS/AD-3.

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